

STA 445 S26 Assignment 5

2026-03-26

Problem 1

Convert the following to date or date/time objects.

```
library(mosaicData)
library(dplyr)
library(lubridate)
library(ggplot2)
```

a. September 13, 2010.

```
as.Date("September 13, 2010", format = "%B %d, %Y")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
as.Date(sub("Sept", "Sep", "Sept 13, 2010"), format = "%b %d, %Y")
```

```
## [1] "2010-09-13"
```

c. Sep 13, 2010.

```
as.Date("Sep 13, 2010", format = "%b %d, %Y")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

This format would not be an acceptable method to use the code because R does not recognize just S. In order to be recognized in R it would need to be either Sep which is %b or September which is %B.

e. 07-Dec-1941.

```
as.Date("07-Dec-1941", format = "%d-%b-%Y")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

The reason for this being wrong is because it can be interpreted in two different ways. It can either be the month then the day or the day and then the month being stated. Since there is not enough information to determine which one it is then it will cause inconsistency and uncertainty in the code.

g. 21-5-1998. Comment on why you know you are correct.

The reason for this being correct is because we can understand that there are no months above 12 and since we have 21 as the first placeholder this means it would need to be the day and not the month. The second value would have to be the month. There is no uncertainty in this date due to there only being one possibility where the month value is supposed to go.

h. 2020-May-5 10:30 am

```
as.POSIXct("2020-May-5 10:30 am", format = "%Y-%b-%d %I:%M %p")
```

```
## [1] "2020-05-05 10:30:00 MST"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```
as.POSIXct("2020-May-5 10:30 am PDT", format = "%Y-%b-%d %I:%M %p", tz = "America/Los_Angeles")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```
as.POSIXct("2020-May-5 10:30 am AST", format = "%Y-%B-%d %I:%M %p", tz = "America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```
birthday <- as.Date("2006-03-16")
birthday_sixty_four <- as.Date(
  paste0(
    as.numeric(
      format(birthday, "%Y")) + 64,
      format(birthday, "-%m-%d")))
```

```
birthday_sixty_four
```

```
## [1] "2070-03-16"
```

b. Calculate your current age (in years).

```
today's_date <- as.Date("2026-03-25")
age <- as.numeric(format(today's_date, "%Y")) -
  as.numeric(format(birthday, "%Y"))

if (format(today's_date, "%m-%d") < (format(birthday, "%m-%d")))
{
  age <- age - 1
}

age
```

```
## [1] 20
```

c. Using your result in part (b), calculate the date of your next birthday.

```
next_birthday <- as.Date(
  paste0(
    as.numeric(format(birthday, "%Y")) + age + 1,
    format(birthday, "-%m-%d")
  ))

next_birthday
```

```
## [1] "2027-03-16"
```

d. The number of *days* until your next birthday.

```
days_next_birthday <- as.numeric(next_birthday - today's_date)
days_next_birthday
```

```
## [1] 356
```

e. The number of *months* and *days* until your next birthday.

```

get_months_days <- function(start_date, end_date)
{
  months <- (as.numeric(format(end_date, "%Y")) - as.numeric(format(start_date, "%Y"))) * 12 +
    (as.numeric(format(end_date, "%m")) - as.numeric(format(start_date, "%m")))

  if (as.numeric(format(end_date, "%d")) < as.numeric(format(start_date, "%d")))
  {
    months <- months - 1
  }

  temp_date <- seq(start_date, by = paste(months, "months"), length.out = 2)[2]

  days <- as.numeric(end_date - temp_date)

  list(months = months, days = days)
}

result <- get_months_days(todays_date, next_birthday)
result$months

```

```
## [1] 11
```

```
result$days
```

```
## [1] 19
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```

call_time <- as.POSIXct("2015-05-08 15:00:00", tz = "America/Phoenix")
time_auk <- format(call_time, tz = "Pacific/Auckland")

time_auk

```

```
## [1] "2015-05-09 10:00:00"
```

The time would be 10 Am on May 9, 2015 ## Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

- a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```
data(Births78)

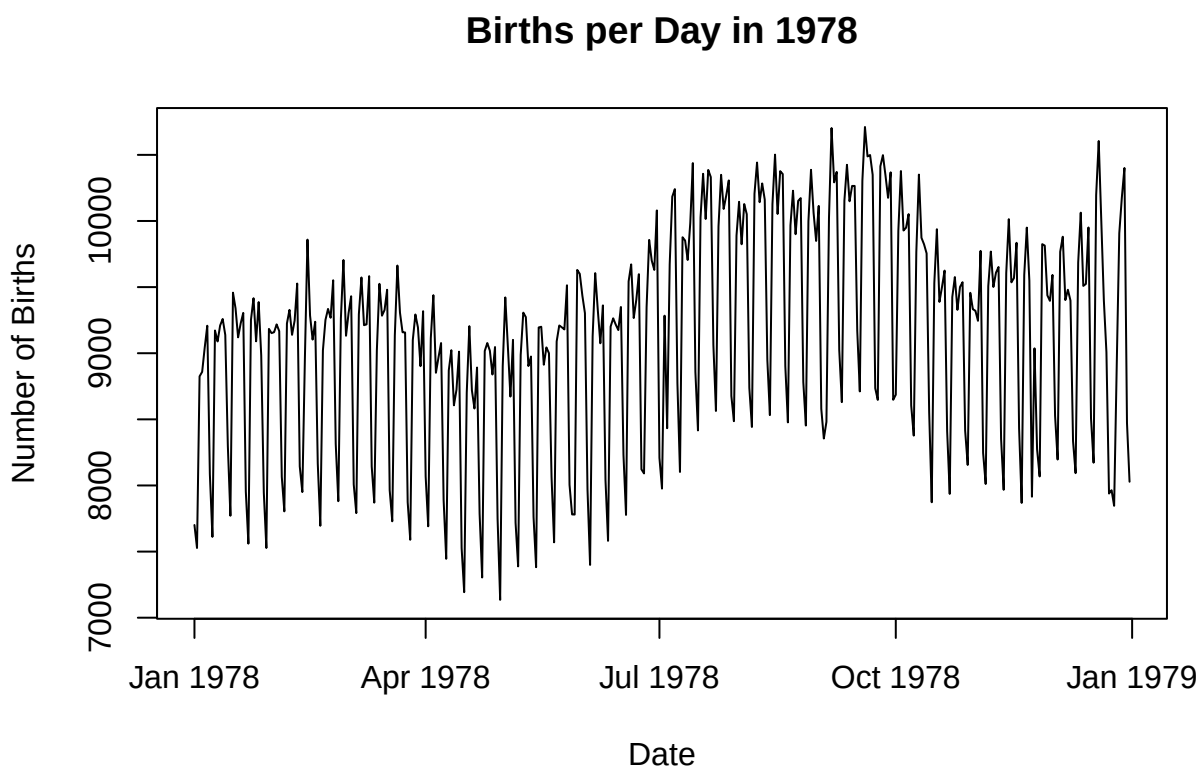
Births78 <- Births78[, c("date", "births")]

head(Births78)
```

```
##      date births
## 1 1978-01-01  7701
## 2 1978-01-02  7527
## 3 1978-01-03  8825
## 4 1978-01-04  8859
## 5 1978-01-05  9043
## 6 1978-01-06  9208
```

- b. Graph the number of `births` vs the `date` with `date` on the x-axis. What stands out to you? Why do you think we have this trend?

```
plot(Births78$date, Births78$births,
     type = "l",
     xlab = "Date",
     ylab = "Number of Births",
     main = "Births per Day in 1978")
```



As seen in the graph, there is a weekly cycle in the number of births, with fewer births occurring on

weekends. This may be partly due to the fact that induced labor and C-sections are more common on weekdays than weekends.

- c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
Births78 <- Births78 %>%  
  mutate(dow = wday(date, label = TRUE, abbr = FALSE))  
  
head(Births78)
```

##		date	births	dow
## 1		1978-01-01	7701	Sunday
## 2		1978-01-02	7527	Monday
## 3		1978-01-03	8825	Tuesday
## 4		1978-01-04	8859	Wednesday
## 5		1978-01-05	9043	Thursday
## 6		1978-01-06	9208	Friday

- d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(Births78, aes(x = date, y = births, color = dow)) +  
  geom_point() +  
  labs(title = "Births per Day in 1978 by Day of Week",  
        x = "Date",  
        y = "Number of Births")
```

Births per Day in 1978 by Day of Week

